

Christie J Purackal

Machine Learning × Geometric Vision | SLAM | Robotics

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Summary

Senior computer vision engineer with 9 years building real-time perception for robotics and embedded products, working at the intersection of deep learning and classical geometric vision: learned models grounded in geometry, plus the systems engineering to ship them on constrained hardware, from consumer devices to quadruped robots.

Core maintainer of Kornia, the open-source **differentiable computer vision** ecosystem (**11K+ GitHub stars, 3M+ monthly downloads**) that puts classical vision operators inside deep-learning pipelines, and creator of **kornia-slam**, Kornia's visual SLAM framework, now a **Google Summer of Code 2026** project I mentor. I go after technically hard problems with real-world impact.

Core Skills

Computer Vision • SLAM • Localisation • Pose Estimation • Calibration • Sensor Fusion • Factor-Graph Optimisation • Real-Time Systems • Deep Learning • Robotics • Python • C++ • Rust • ROS2 • ONNX • CUDA • Agile (Scrum/Kanban)

Open Source

2022–present *Kornia: core member/maintainer*

- Core maintainer of **Kornia**, an open-source computer-vision ecosystem with **11K+ GitHub stars, 3M+ monthly PyPI downloads**, and **100M+ total PyPI downloads**.
- Creator of **kornia-slam**, Kornia's modular open-source **multi-sensor SLAM** framework in Rust for real-time pose estimation and mapping. Architected and built its estimation core, supporting monocular, monocular-inertial, stereo, and stereo-inertial odometry through a common tightly-coupled optimisation backend.
- Maintain **kornia-rs**, Kornia's Rust computer-vision library for edge CV. Primary author of **kornia-algebra** and **kornia-3d**, covering Lie groups, factor-graph optimisation, and 3D geometry: pose estimation, point-cloud registration, etc.
- Maintain **kornia-pytorch**, Kornia's main PyTorch-based computer-vision library for differentiable image processing, geometry, augmentation, and deep-learning workflows.

2025–present *Google Summer of Code, Org Administrator & Mentor*

- Organisation Administrator and Mentor for Kornia in **GSoC 2025 and 2026**, leading programme execution alongside the project founder; mentored visual-fiducial and feature detection (AprilTag, FAST, ORB) in 2025, and **SLAM (kornia-slam)** and GPU acceleration for kornia-rs in 2026.
- Represented Kornia at the **GSoC Mentor Summit 2025** (Munich), alongside delegates from CERN, DeepMind, Git, and Python.

Work Experience

2023–present *Machine Learning Engineer, Audioscenic, Southampton, United Kingdom*

- Designed a novel multi-task CNN that predicts heatmap localisation, 2D keypoint regression, and direct continuous 6D rotation regression from a single RGB image in one edge-deployable network, feeding PnP-based 3D pose recovery and reducing keypoint jitter for more stable head tracking.
- Replaced the legacy two-stage ear-tracking pipeline with a single-stage CNN, cutting memory 100× and compute 20% for real-time inference on low-power x86/ARM SoCs.
- Built the internal ML platform supporting this work (dataset management, experiment tracking, evaluation, and ONNX/MNN export) for reproducible model development.
- Planned and delivered this work within Scrum/Kanban processes, translating product requirements into user stories and detailed engineering specifications.
- Deployed these models through a graph-based C++ inference runtime with ONNX/MNN backends, spanning heterogeneous targets: laptops, monitors, soundbars, CPUs, NPUs, and ARM SoCs. Shipped in award-winning consumer products including the Dell 32 Plus 4K QD-OLED Monitor, Lenovo laptops, and a Razer soundbar.

2019–2023	Robotics / SLAM Consultant (Remote), SIW Engineering, Ecosys Group, Penang, Malaysia - Implemented a dense visual-inertial SLAM pipeline in ROS2 for an MIT Mini Cheetah-style quadruped, combining factor-graph state estimation (GTSAM) with loop closure and pose-graph optimisation for globally consistent trajectories, and CUDA-accelerated dense mapping, all running on-board an NVIDIA Jetson AGX Xavier. - Built calibration and sensor-alignment modules for multi-sensor perception: camera intrinsic validation, extrinsic estimation, and IMU-camera alignment (Kalibr). - Integrated semantic-segmentation networks into the ROS2 perception stack for scene understanding and mapping robustness in warehouse-scale environments. - Developed tooling for calibration checks, trajectory inspection, and post-run analysis to debug localisation drift, sensor-alignment issues, and mapping failures.
2018–2019	Team Lead, ML / Computer Vision, Agrima Infotech, Kochi, India Led a 4-engineer ML team building production object-detection and segmentation models for a large-scale food-recognition system, scaling the curated dataset to 100K+ annotated images with a human-in-the-loop labelling pipeline. Built Kubernetes-based training infrastructure for distributed training and faster team iteration. Technology later acquired by BigBasket.
2017–2018	Junior Software Engineer, Lithion Power, Kochi, India Implemented prototype modules for lane detection, semantic road segmentation, and vehicle detection as part of an early autonomous-driving perception stack. Built a driver-monitoring system using facial-behaviour cues to identify distraction and drowsiness.

Education

2013–2017	Cochin University of Science and Technology, B.Tech, Information Technology
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Projects & Initiatives

2018–2019	Kerala Artificial Intelligence Initiative, Co-founder Open community supporting AI research and collaboration in Kerala.
2013–2017	Bitverse, Co-founder Built and launched multiple games across Android and Windows platforms, including releases on the Google Play Store (10K+ downloads).

Additional

UK Skilled Worker visa | open to on-site work across the UK